

Section 4D. Maximum Likelihood

Victor M. Preciado, PhD MIT EECS
Dept of Electrical & Systems Engineering
University of Pennsylvania
preciado@seas.upenn.edu

Maximum Likelihood Estimation (MLE)

Given a training dataset $\mathcal{D}_{\text{Tr}} = \{(x_i, y_i)\}_{i=1}^N$ of independent samples with $y_i \in \{0, 1\}$, we would like to estimate the parameters β_0 and β_1 in the logistic function that best explain the data

- ▶ **Maximum likelihood criterion:** We can use a criterion, called *maximum likelihood estimation* (MLE), to estimate the values of the parameters β_0 and β_1 from $\mathcal{D}_{\text{Tr}}$
- ▶ Assume that the input variables $\{x_i\}_{i=1}^n$ are given. The *likelihood function* is defined as the conditional probability of observing a set of outputs $\{y_i\}_{i=1}^n$ given the inputs $\{x_i\}_{i=1}^n$, i.e.,

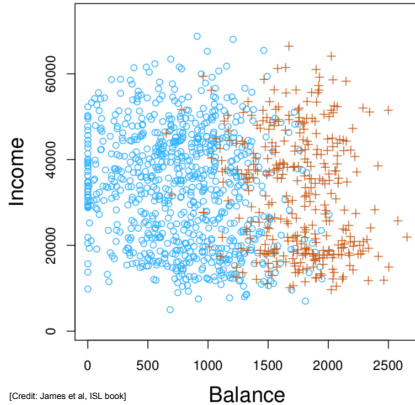
$$\begin{aligned}\ell(\theta_0, \theta_1) &= \prod_{i=1}^N \Pr(Y = y_i | X = x_i; \theta_0, \theta_1) \\ &= \prod_{i: y_i=1} p_1(x_i; \theta_0, \theta_1) \prod_{i: y_i=0} p_0(x_i; \theta_0, \theta_1)\end{aligned}$$

- ▶ We can estimate the unknown parameters using the MLE criterion:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \max_{\theta_0, \theta_1} \ell(\theta_0, \theta_1)$$

MLE: Numerical Example

Consider the Default dataset in the figure below. Using Balance as the only input variable find the coefficients of a univariate logistic curve (as well as their standard deviations)



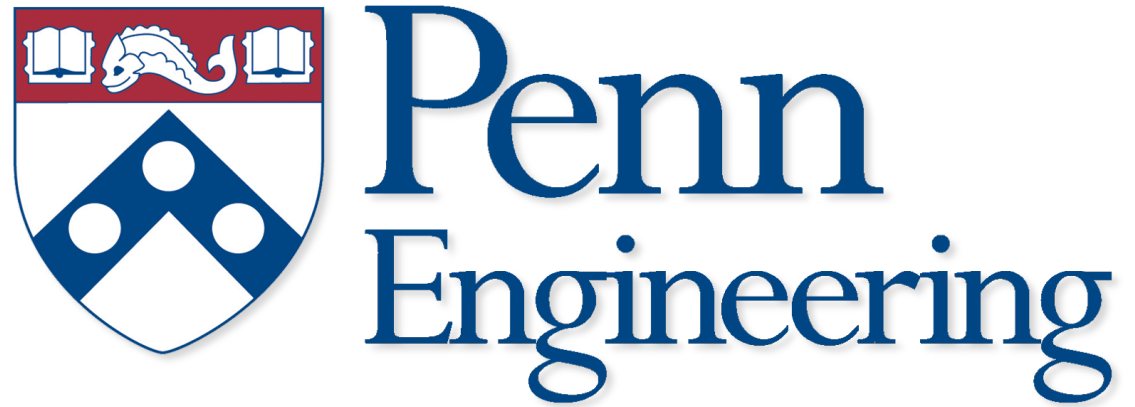
[Credit: James et al, ISL book]	Coefficient	Std. Error	Z-statistic
Intercept	-10.6513	0.3612	-29.5
balance	0.0055	0.0002	24.9

Maximum Likelihood: Making Predictions

- ▶ Once we find $\hat{\beta}_0$ and $\hat{\beta}_1$, we can make predictions about the probability of an individual being in the **Default** class when we know her **Balance**:

$$\hat{p}_1(\text{Balance}) = p_1(\text{Balance}; \hat{\beta}_0, \hat{\beta}_1) = \frac{e^{-10.65 + 0.0055 \times \text{Balance}}}{1 + e^{-10.65 + 0.0055 \times \text{Balance}}}$$

- ▶ For example, an individual with a Balance=\$1000 has a probability of defaulting of 0.006 (0.6%)
- ▶ For example, an individual with a Balance=\$2000 has a probability of defaulting of 0.586 (58.6%)



Copyright 2020 University of Pennsylvania
No reproduction or distribution without permission.